

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular & Supplementary Summer 2024 Course: B. Tech. Branch: Common to All Branches Semester: II Subject Code & Name: Engineering Chemistry BTBS202 Max Marks: 60 Date: 14/06/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/ CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain Zeolite process for softening of water with its advantages and Disadvantages.	2	6
B)	Define- 1) Hard water 2) Soft water 3) Scale 4) Sludge 5) Temporary hardness 6) Permanent hardness	1	6
C)	Describe the determination of dissolved oxygen in water by Iodometric method.	2	6
Q.2	Solve Any Two of the following.		12
A)	State Gibb's Phase rule equation. Explain the term number of phase and degree of freedom with examples.	1	6
B)	Explain the Phase diagram of one component water system.	2	6
C)	Explain Silver – Lead two component alloy system with phase diagram.	2	6
Q. 3	Solve Any Two of the following.		12
A)	Explain Hydrogen evolution mechanism of wet corrosion.	2	6
B)	Explain factors influencing rate of corrosion.	1	6
C)	Write a note on cathodic protection method of corrosion.	2	6
Q.4	Solve Any Two of the following.		12
A)	Explain how the percentage of Carbon, Hydrogen and Nitrogen in coal can be estimated.	2	6
B)	Explain liquid lubricants in detail.	1	6
C)	Explain in detail Refining of petroleum.	2	6
Q. 5	Solve Any Two of the following.		12
A)	Explain the principle of Conductometric titration with example of a) Strong acid and Strong base b) Weak acid and Weak base	2	6
B)	Explain Ostwald's theory of acid – base indicator.	2	6
C)	Define the terms specific conductance, molecular conductance, equivalence conductance and cell constant with their units.	1	6
	*** End ***		